

BARCELONA, SPAIN • ARCHITECTURAL MONOGRAPH



BIMCO

VOLUME II: RESIDENTIAL APARTMENTS & FACADES

THE HIGH-PERFORMANCE BUILDING ENVELOPE & COLLECTIVE LIVING

Curated Monograph of Urban Mid-Rise, Terraced Enclaves & High-Rise Facades

COLLABORATIVE LEADERSHIP & PRINCIPALS

SOHEIL MASTI & SIAVASH PAZOOKI

BIM Director & Computational Architect | Senior Architectural Designer & Visualizer

BIMCO STUDIO • BARCELONA, SPAIN

BARCELONA, SPAIN • EDITION 2026

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VERIFIED BIMCO SHEET

BCN

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SPAIN

THE URBAN FACADE & COLLECTIVE FORM

Tectonic Depth, Material Permanence, and Solar Modulation

The facade is the public face of architecture—the mediator between the urban realm and private human dwelling. At BIMCO Studio Barcelona, we treat the building skin as a high-performance tectonic system that balances light, privacy, acoustic buffering, and energy conservation.

Our residential projects—from terraced mountain dwellings in Dalkhani to mid-rise luxury enclaves in Darrous and high-density modular towers—demonstrate rigorous structural logic. Every joint, reveal, and stone anchor is coordinated in BIM to bridge the gap between bold aesthetic visions and flawless on-site installation.

APARTMENTS & FACADE INDEX

Curated Sequence from Bespoke Facades to High-Rise Towers

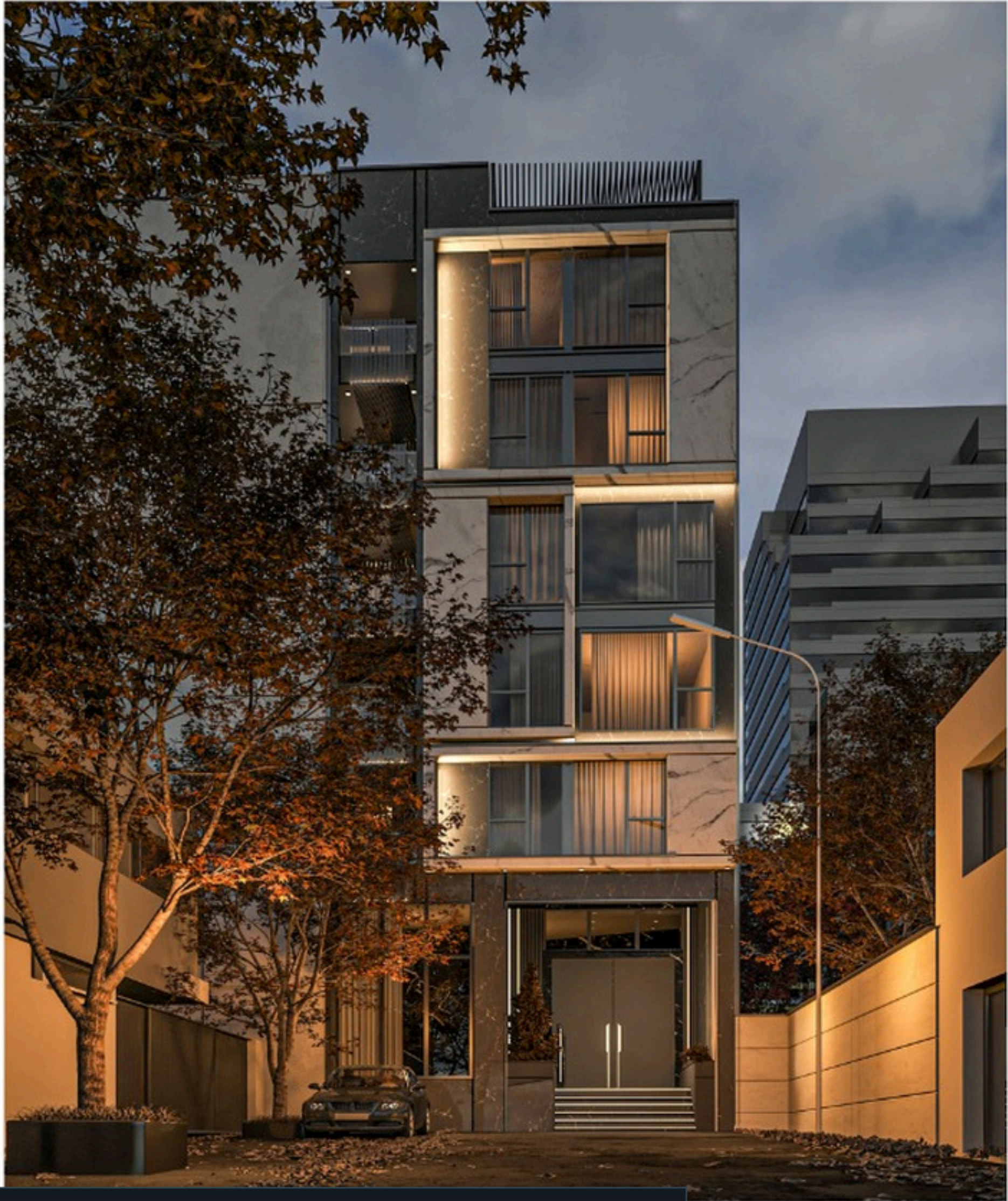
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02	DARROUS NIGHT ILLUMINATION	Photometric Studies & Nocturnal Depth	PGS 06-07
03	DARROUS ELEVATION GEOMETRY	Vertical Louvers & Solar Angle Optimization	PGS 08-09
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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

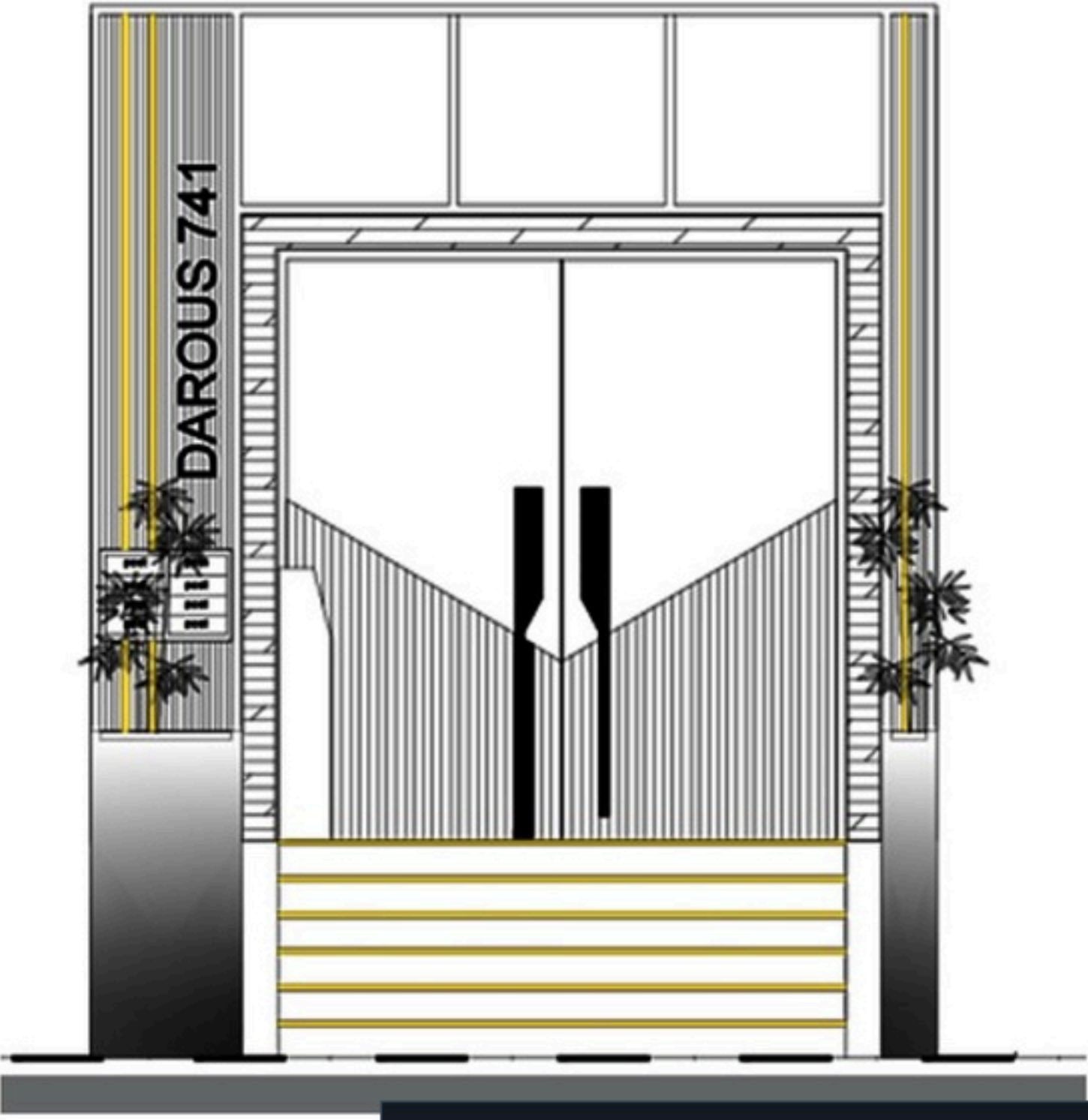
From Barcelona, BIMCO approaches multi-family residential architecture through high-p...

BIM & STRUCTURAL INTEGRITY SPECS

Revit Facade Families LOD 350, Thermal Bridging Mitigation, Ventilated Stone Lou...



A front elevation design is basically a part of as scenic design which highlights the scenic element or the entire set as seen from the front view with or without measurements. Most of the front elevation designs shows the measurements to get a better understanding of the building so that it can be drafted or quickly drawn by an architect. The front elevation of a home plan is a straight-on view of the house as if you were looking at it from a perfectly centered spot on the same plane as the house.

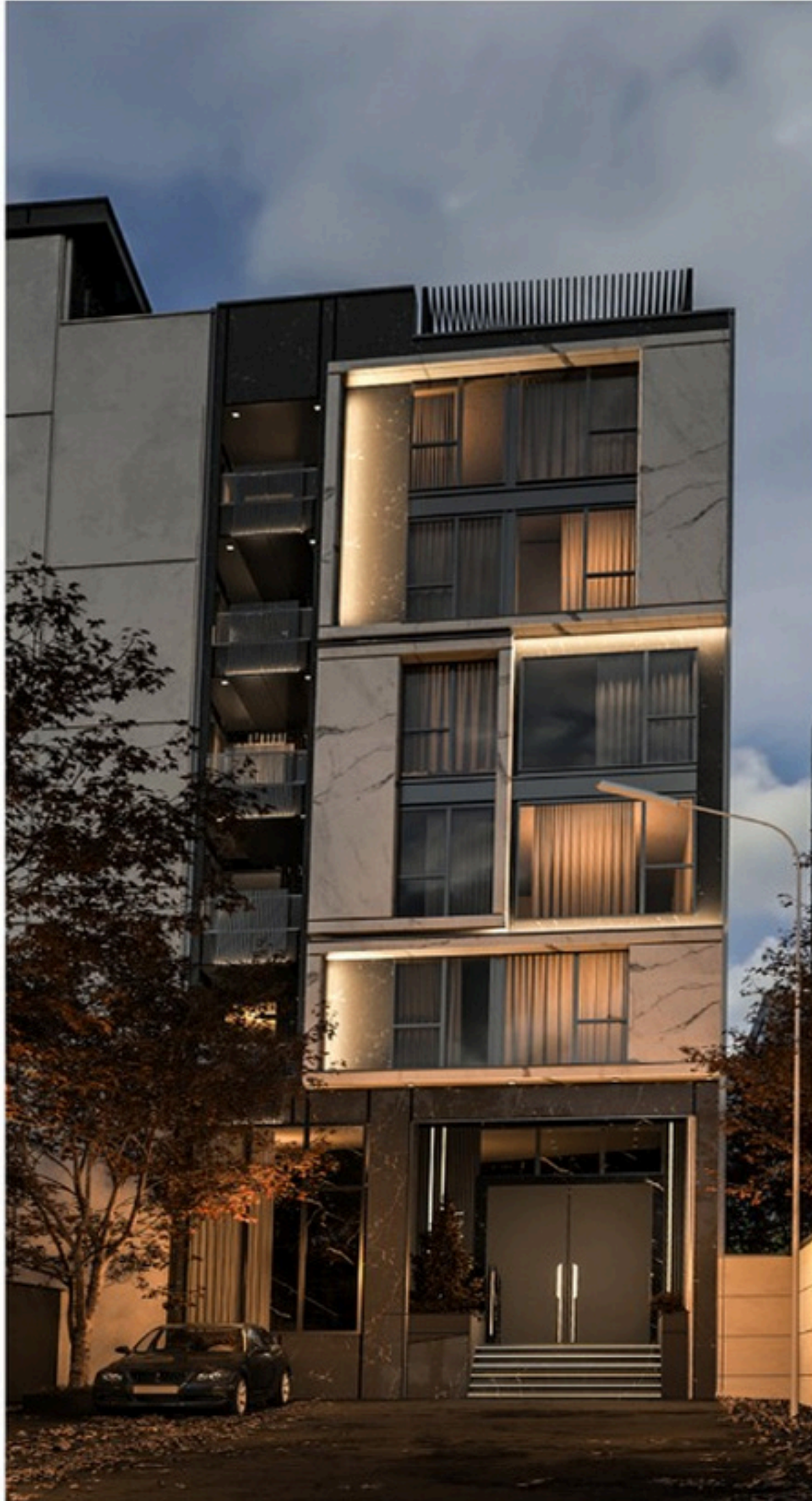


ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Luxury residential entrance foyer in Darrous district featuring travertine masonry, b...

BIM & STRUCTURAL INTEGRITY SPECS

Ventilated facade substructure, stainless steel brackets, concealed mechanical j...



Modern house elevations are distinguished by several defining features

Minimalism: Modern designs often emphasize a minimalist approach, characterized by clean, uncluttered lines and a lack of ornamental detailing

Large Windows: Expansive windows are a hallmark of modern elevations, allowing for ample natural light and a seamless connection between the indoors and outdoors

Innovative Materials: The use of innovative materials such as glass, steel, and concrete is common, offering both aesthetic appeal and functional benefits

Geometric Forms: Modern elevations frequently incorporate bold geometric forms and asymmetrical compositions, making strong architectural statements

Integration with Nature: Many modern designs focus on harmonizing the building with its natural surroundings, often using elevation to enhance this connection



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Nocturnal illumination design integrated into stone reveals and window frames, emphas...

BIM & STRUCTURAL INTEGRITY SPECS

Warm 3000K LED linear wall grazing, anti-glare baffles, IP67 waterproof rated fi...



Modern house elevation refers to the external design and architectural elements of a house's front façade. It plays a crucial role in defining the aesthetic appeal and character of a modern home. In contemporary architecture, house elevation is not just about creating a visually appealing structure; it's about blending form with functionality, innovation with sustainability, and design with the surrounding environment

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

South-facing facade geometry utilizing vertical louver elements to regulate solar gai...



Interior elevations are very important to the owner and the builder. Interior elevations are usually drawn for cabinets and special built-in fixtures. They normally show everything that will be located on the finished wall. The set of construction drawing includes interior elevations of the teller's counter. These elevations, are drawn in conjunction with the floor plan so that all dimensions for constructing the cabinets are shown on one of the two drawings.



BIM & STRUCTURAL INTEGRITY SPECS

Parametric shadow calculation, extruded aluminum profiles, dual low-E insulated ...



An elevation drawing is an orthographic projection drawing that shows one side of the house. The purpose of an elevation drawing is to show the finished appearance of a given side of the house and furnish vertical height dimensions. Four elevations are customarily drawn, one for each side of the house.



Lighting Legend

BIM & STRUCTURAL INTEGRITY SPECS

Revit LOD 350 facade details, seismic movement joints, air barrier continuity, d...

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Executive construction drawings with 1:20 joinery, stone reveal brackets, thermal fla...



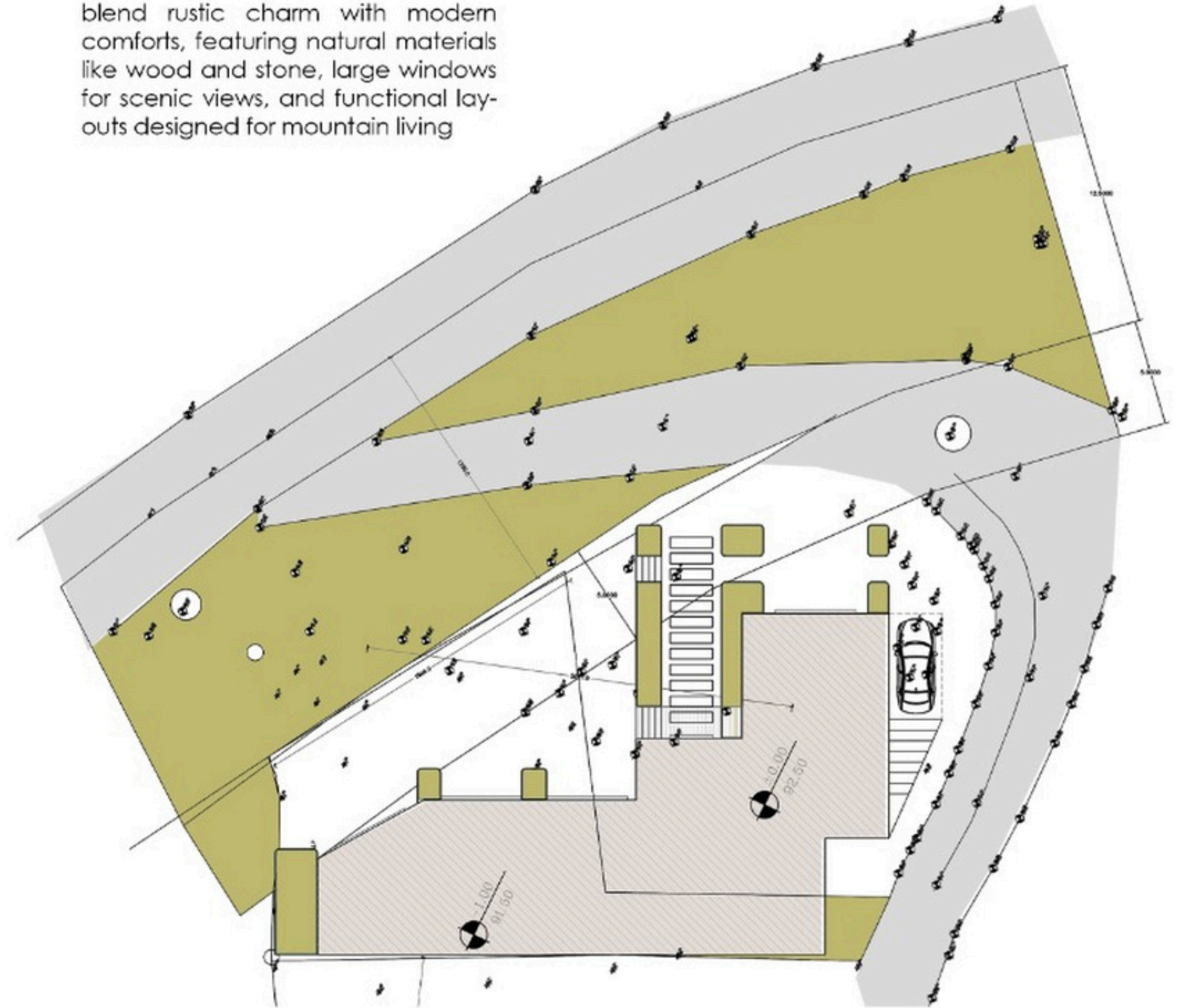
RESIDENTIAL DALKHANI PROJECT



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Multi-family stepped collective housing contoured along steep mountainsides, providin...

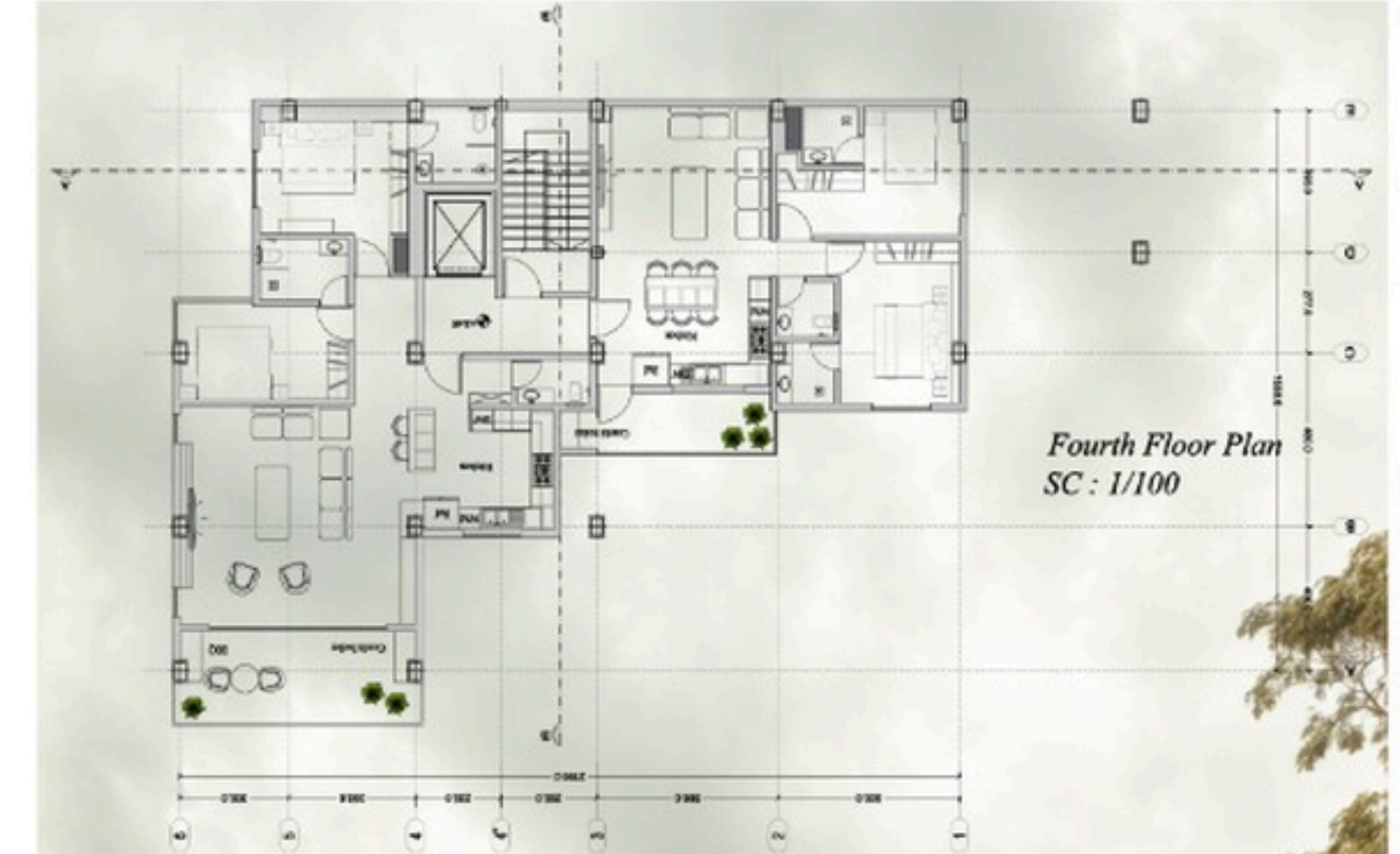
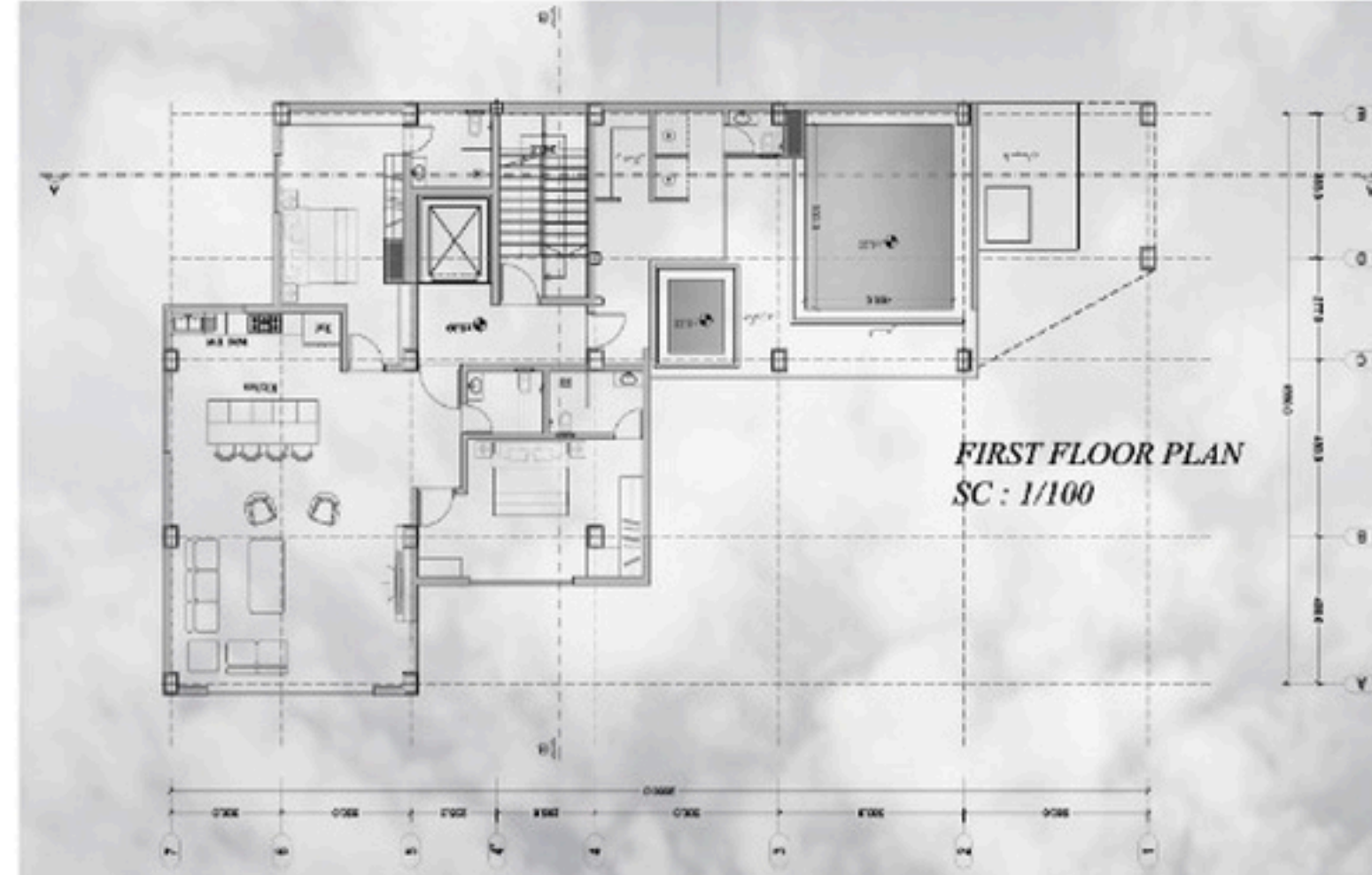
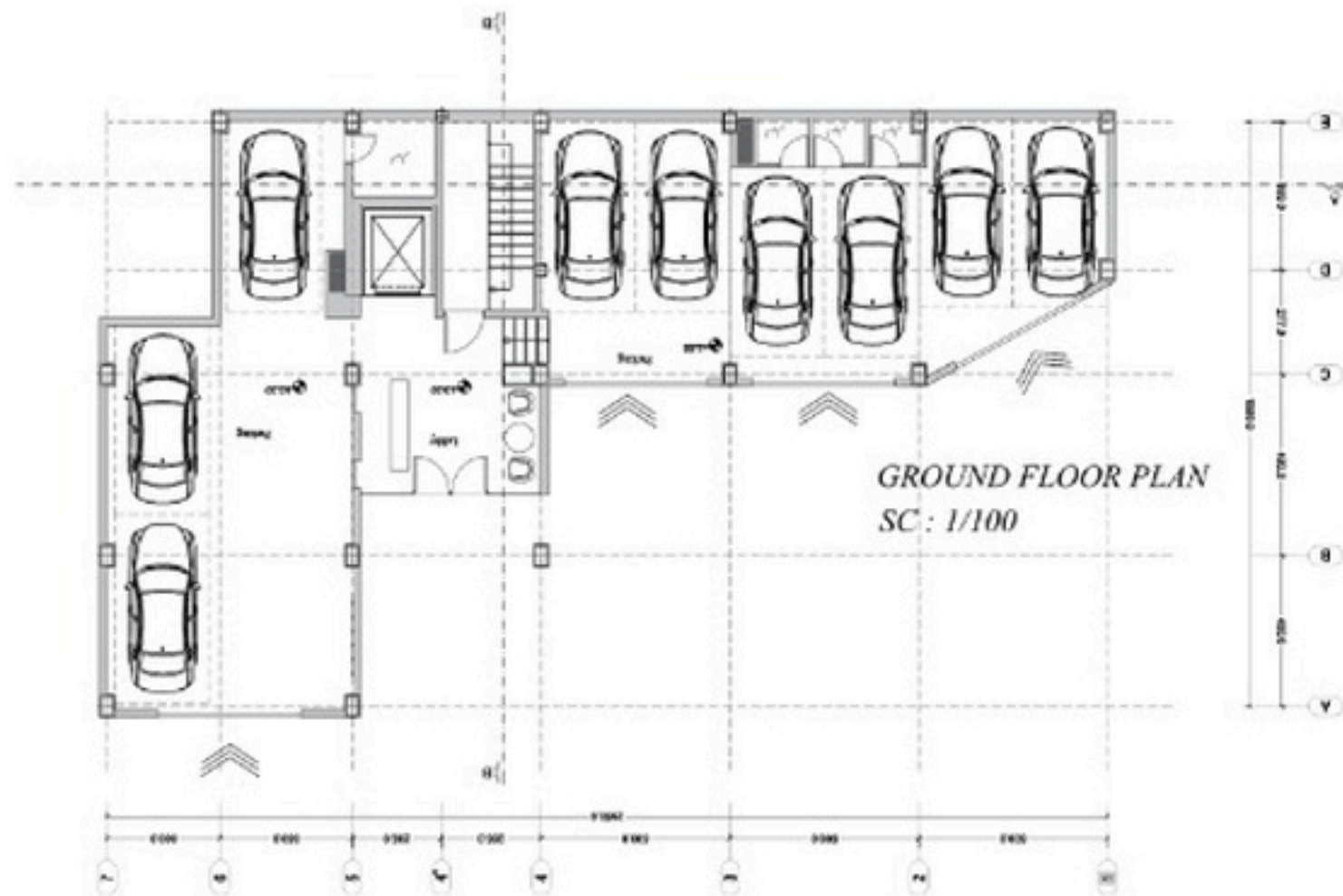
Residential mountain homes often blend rustic charm with modern comforts, featuring natural materials like wood and stone, large windows for scenic views, and functional layouts designed for mountain living



محل قرار گیری ساختمان در سایت پلان

BIM & STRUCTURAL INTEGRITY SPECS

Terraced concrete foundation, soil-nail retaining structures, green roof water r...



:Natural Materials

Wood and stone are prevalent, often with exposed beams and rustic details

:Large Windows

Expansive windows are a hallmark, designed to maximize views of the surrounding mountains and landscape

:Outdoor Living Spaces

Front and back porches, patios, and outdoor fireplaces are common features

:Open Floor Plans

Large, open living areas are typical, often complemented by more private bedrooms and functional spaces like mudrooms



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Contoured architectural floor plans stepping up the terrain with private entries, nat...

BIM & STRUCTURAL INTEGRITY SPECS

Stepped shear-wall grid, dual-aspect ventilation, acoustic party-wall assemblies...



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Longitudinal building section demonstrating balcony overhangs, daylight penetration, ...



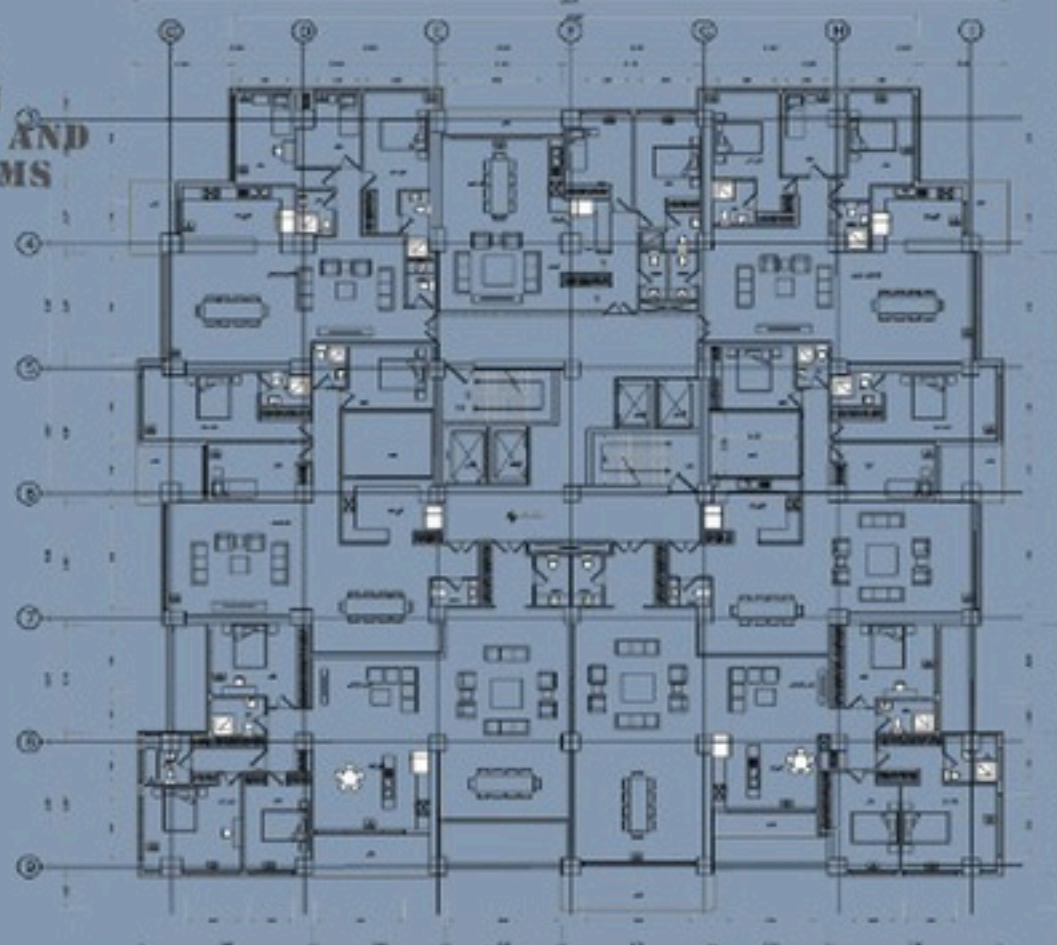
BIM & STRUCTURAL INTEGRITY SPECS

Cascading structural calculations, inverted roof waterproofing, exterior grade t...



ELEVATION DESIGN THE EXTERIOR DESIGN NAME: ELEVATION AND PLAN DESIGN MR SHAMS YEAR: 2018

Emphasis is important in the elevation design to lead the eye of the viewer to the entrance, for example, or other important parts, and there are many ways to achieve emphasis.



پلان تپ طبقات ۱۰ و ۷ و ۳ و ۲
مقیاس ۱:۱۰۰



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Urban mid-rise residential architecture with deep cantilevered green balconies, integ...



While your architectural elevation needs to have unity it, also, needs variety or it will be boring. On first glance, both concepts might seem to be opposing, but in design, this is what "we call" variety within unity



First of all, we have to be aware that architectural elevation design is like any other design in general, meaning that the basic rules and principles of design apply. While we may disagree on the word "rules" in design, but there a set of characteristics that are always found in a successful design, whether they were respected consciously or subconsciously



BIM & STRUCTURAL INTEGRITY SPECS

Automated micro-drip planter irrigation, structural cantilever load calculations...



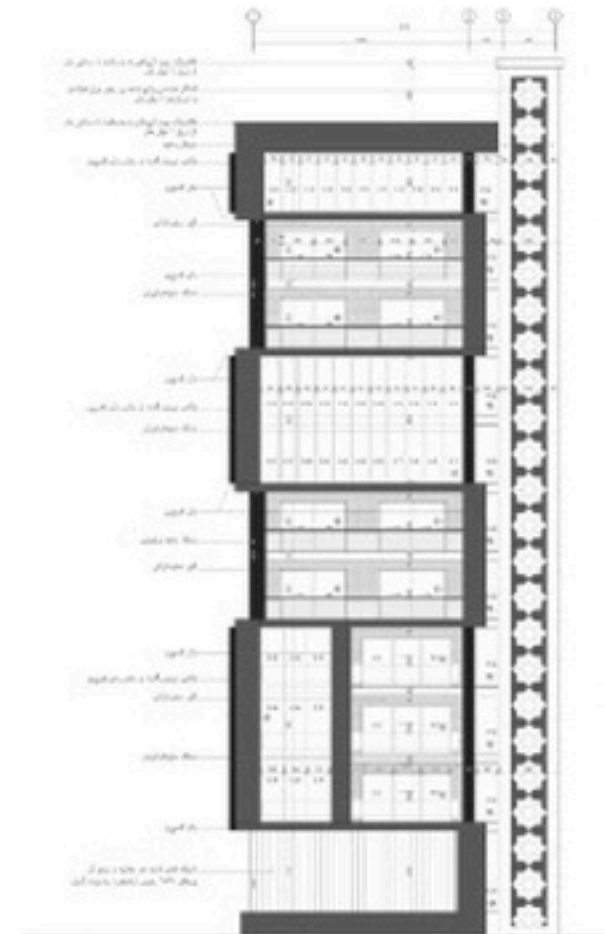
EXTERIOR DESIGN MODERN ELEVATION THE EXTERIOR DESIGN

Proportions are vital in architecture, and naturally in elevation design. They can make or break a design. Rules were being devised since the time of ancient Greece to obtain perfect proportions, and here are some of the most commonly used proportions in design



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Rhythmic facade articulation balancing textured masonry with full-height floor-to-ceiling glass.



Emphasis is important in the elevation design to lead the eye of the viewer to the entrance, for example, or other important parts, and there are many ways to achieve emphasis. Emphasis, by contrast, is one way. This contrast can be created by color or shape and texture.

There is also emphasis by isolation. Isolating one element from the successive pattern of the rest makes it a focal point in a way.



BIM & STRUCTURAL INTEGRITY SPECS

Thermal break structural brackets, laminated acoustic balustrades, integrated hi...

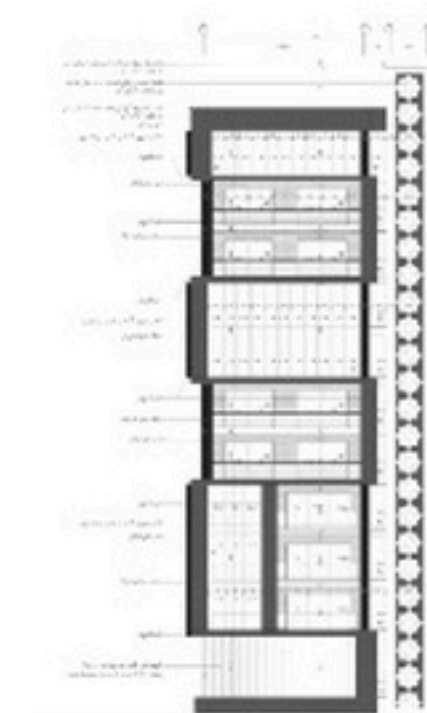


EXTERIOR DESIGN MODERN ELEVATION

THE EXTERIOR DESIGN

Proportions are vital in architecture, and naturally in elevation design. They can make or break a design. Rules were being devised since the time of ancient Greece to obtain perfect proportions, and here are some of the most commonly used proportions in design.

Balance is another essential factor that should be present in the design of an architectural elevation. The most obvious way to achieve balance is by symmetry. This how it has been done since the ancient times. However, symmetry is not the "thing" now. Contemporary architects believe in the more indirect methods.



Emphasis is important in the elevation design to lead the eye of the viewer to the entrance, for example, or other important parts, and there are many ways to achieve emphasis. Emphasis, by contrast, is one way. This contrast can be created by color or shape and texture.

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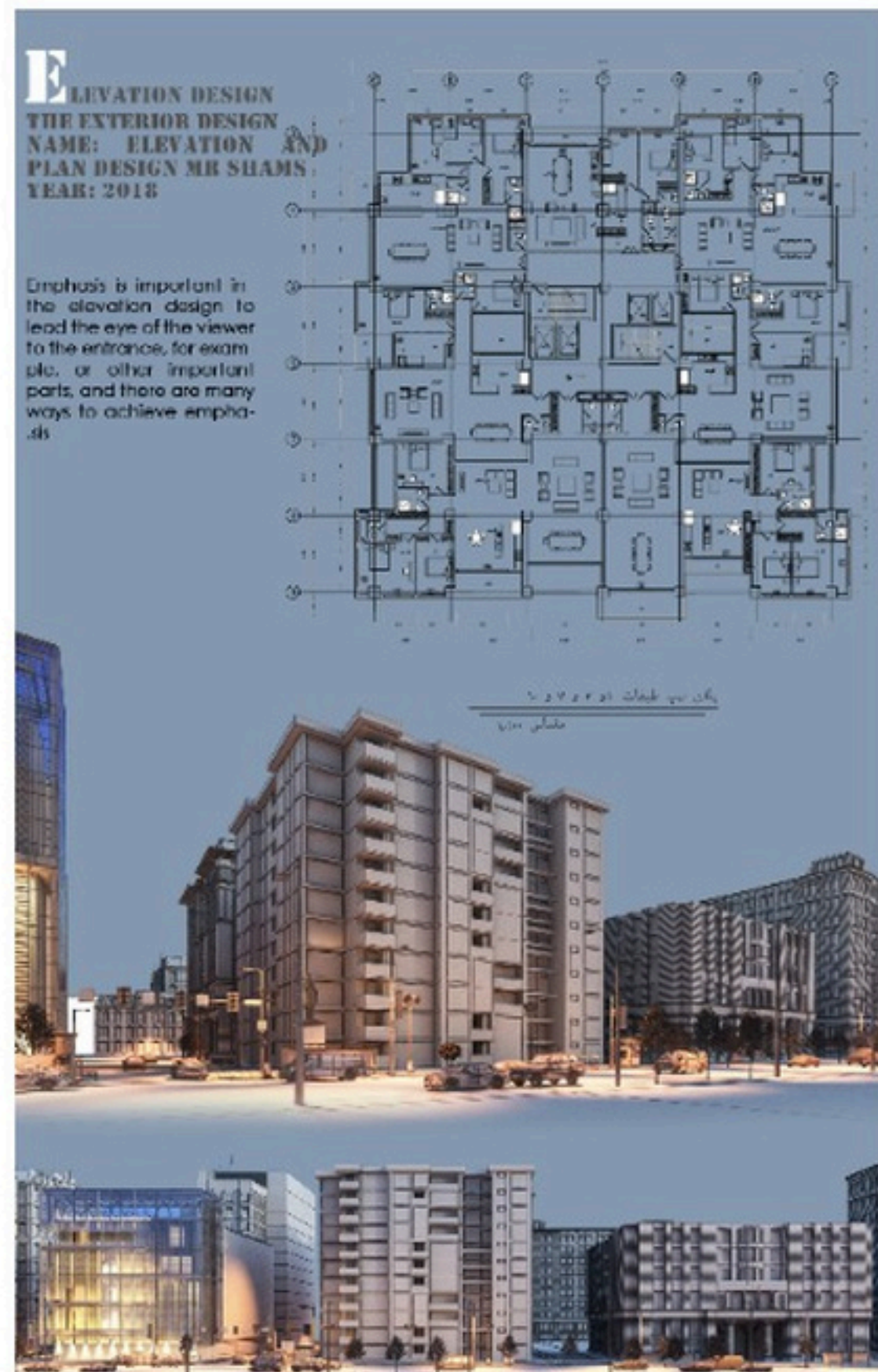


ARCHITECTURAL BRIEF & SPATIAL CONCEPT

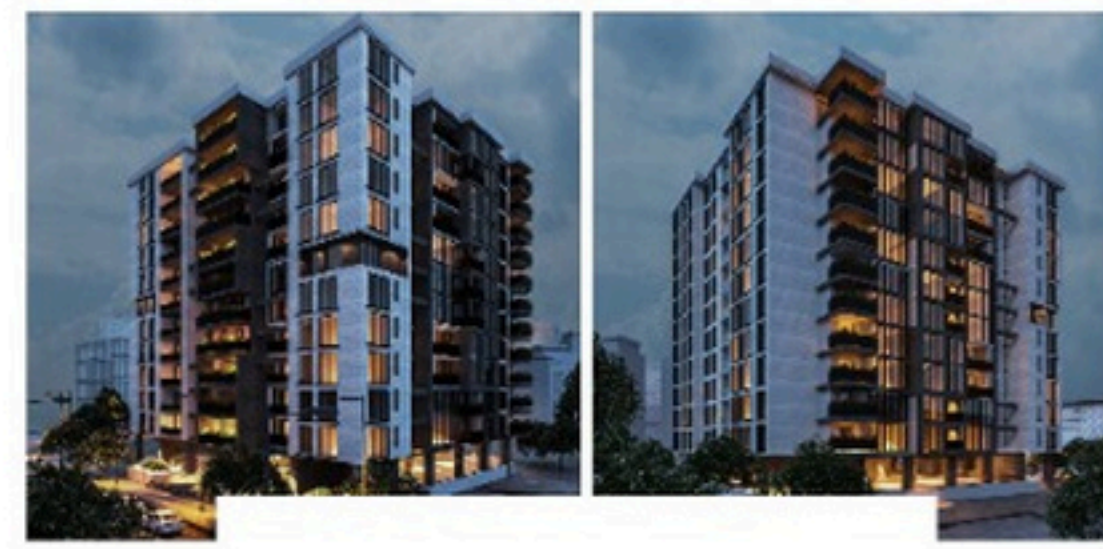
Architectural tower facade modularity exploring repetitive prefabricated panels with ...

BIM & STRUCTURAL INTEGRITY SPECS

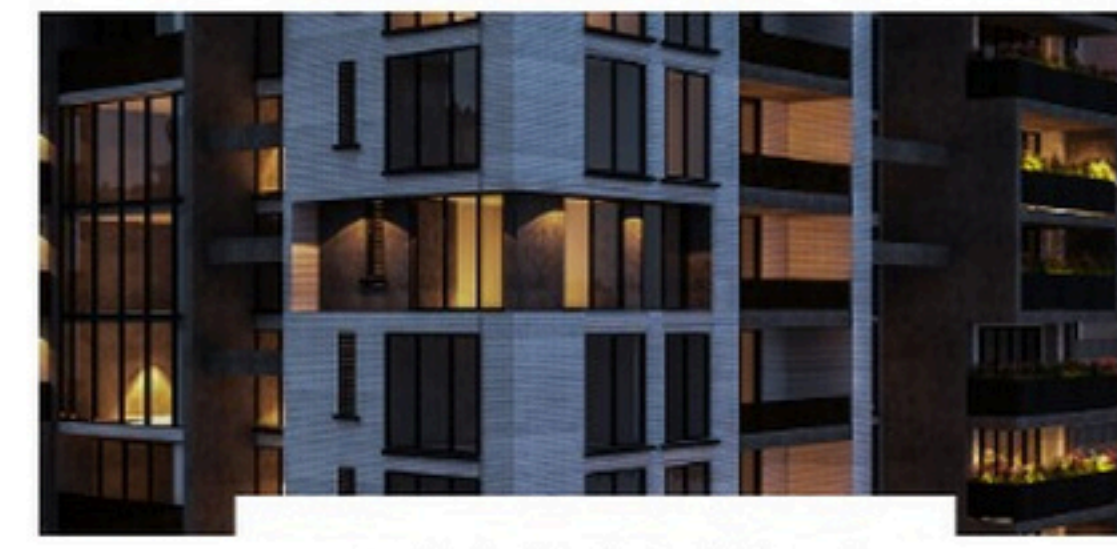
GFRC lightweight composite panels, unitized facade crane installation sequence, ...



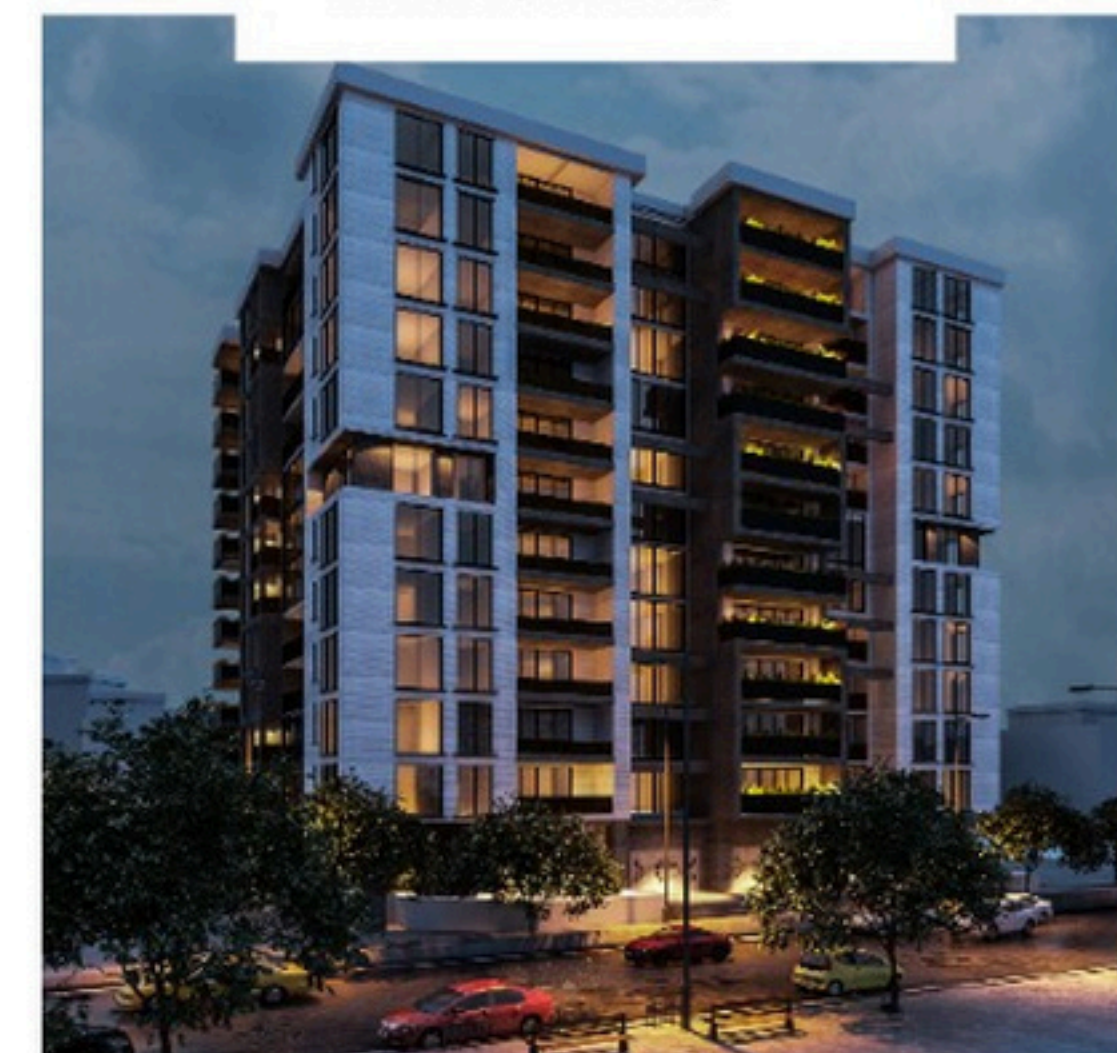
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Your architectural elevation should be harmonious with a degree of unity. Unity makes the different elements and components of the elevation seem to be one, a whole instead of parts. There are different ways to achieve unity





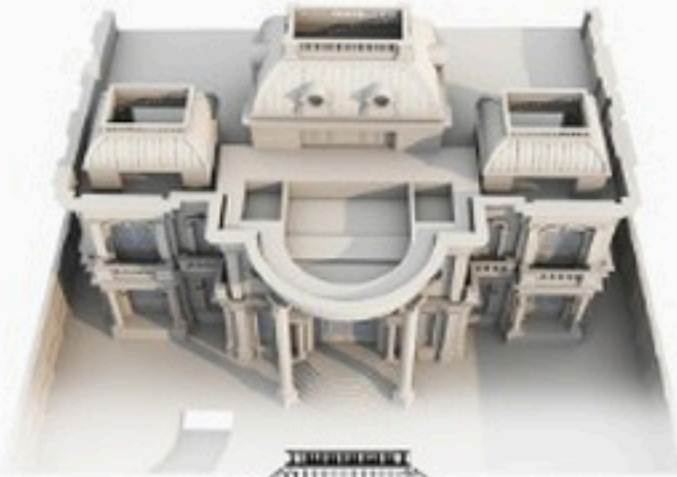
CLASSIC VILLA DESIGN
THE EXTERIOR DESIGN

In classical architecture, colors are limited because only warm colors such as brown, gold, bronze, ... are selected. The most important point that should be observed in choosing the color of such buildings is the use of one to two colors, not more. Now the classic façade is run by various builders and architects for different buildings.



The main and basic color of this façade design should be white, because the stones used to make this type of façade are often bright. We should add that in order to implement this façade model, there is a need to use high-quality stones that can also be cut.
- Frilled columns should not be used and columns with circular sections should be used.

- Along with other columns, small and decorative columns were also used.
- Use the heads of columns that have ancient Roman design.
- Use large-sized arches on which prominent and delicate roles are cared for.
- Use different types of window frames, inscriptions, piers, crests, etc.



Beautiful glass doors and unique façade are the most important positive features of this material. Because of transparency, it helps to disseminate natural light in the lobby environment of the building. Also, to create a suitable insulation, the door can be made with double glass panels (two layers).



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Monumental residential palace exhibiting classical Roman and Neoclassical orders, gra...

BIM & STRUCTURAL INTEGRITY SPECS

Natural limestone masonry, structural entablature tie-backs, carved stone comic...

BIMCO PROTOCOL: DESIGN & MODELING

Stages 01 – 03 • Facade Engineering Protocol

STAGE 1: SOLAR & WIND MICROCLIMATE

Parametric shadow calculations, wind pressure CFD, seasonal solar radiation mapping.

STAGE 2: TECTONIC FACADE MODULATION

Fenestration rhythms, louver profiles, spandrel glass vs. opaque stone balance.

STAGE 3: REVIT LOD 350 FACADE FAMILIES

BIM cladding assemblies, bracket coordination, thermal break insulation modeling.

BIMCO PROTOCOL: VISUALIZATION & BUILD

Stages 04 – 06 • From CGI to On-Site Reality

STAGE 4: DAY & NIGHT PHOTOMETRIC CGI

Cinematic renders in Corona, realistic nighttime illumination, glare evaluation.

STAGE 5: 1:20 CLADDING & SHOP DRAWINGS

Executive facade detailing, anchor bolts, weep holes, thermal bridge mitigation.

STAGE 6: WORKSHOP MOCKUPS & SITE AUDIT

Quality assurance on stone fabrication, unitized panel installation, seal testing.

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

BIMCO Studio Barcelona mandates complete continuity from parametric BIM LOD 350 to si...

BIM & STRUCTURAL INTEGRITY SPECS

ISO 19650 BIM Standards, Autodesk Revit LOD 350, Clash Zero Tolerance, Worksite ...

SOHEIL MASTI

BIM Director & Computational Architecture Lead

BIMCO Studio Barcelona • Spain & International

BIM MANAGEMENT & INFORMATION ARCHITECTURE:

- Strategic BIM implementation, Revit LOD 350 parametric modeling
- Multi-discipline clash audits between architectural form, structural frames, and MEP systems
- Automated quantity takeoffs and material schedules for bespoke builds
- International project delivery across Spain, Europe & Middle East
- Digital construction workflows ensuring zero discrepancies on site

SIAVASH PAZOOKI

Senior Architectural Designer & High-End 3D Visualizer

M.Arch & B.Arch • BIMCO Studio Barcelona Partner

CONCEPT DESIGN & VISUALIZATION LEADERSHIP:

- Lead Architectural Concept Designer for signature luxury commissions & masterplans
- Photorealistic architectural rendering, natural lighting moods & cinematic camera angles
- Topographical terrain modeling, landscape synthesis, and material detailing
- Turnkey executive shop drawings, joinery specifications & on-site build supervision
- 8+ years leading award-winning commissions across residential & urban typologies

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

BIMCO Studio Barcelona unites high-level BIM information architecture with signature ...

BIM & STRUCTURAL INTEGRITY SPECS

Registered Architects, BIM Managers (Autodesk Certified), LOD 350 Audit Speciali...



BIMCO

VOLUME II: RESIDENTIAL APARTMENTS & FACADES

BARCELONA, SPAIN

STUDIO HEADQUARTERS: Barcelona, Spain

DIRECTORS: Soheil Masti & Siavash Pazooki

TELEPHONE & WHATSAPP: +34 610 855 434

OFFICIAL WEB PORTAL: WWW.BIMCO.ES

SERVICES: Architecture • BIM LOD 350 • CGI • Site Supervision